

Ex. 3

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```
print(uuid::UUIDgenerate())
```

```
## [1] "15a9d2fe-4dd2-448d-8d1a-73ef802f5af9"
```

Question 1

```
rent <- c(12500, 13800, 11200, 15000, 12500, 26000,  
          14300, 13100, 12500, 16700, 14900, 13600)  
mean(rent)
```

```
## [1] 14675
```

```
median(rent)
```

```
## [1] 13700
```

```
table(rent)
```

```
## rent  
## 11200 12500 13100 13600 13800 14300 14900 15000 16700 26000  
##      1      3      1      1      1      1      1      1      1      1
```

Question 2

```
rent2 <- rent[-6]  
mean(rent2)
```

```
## [1] 13645.45
```

```
median(rent2)
```

```
## [1] 13600
```

```
table(rent2)
```

```
## rent2  
## 11200 12500 13100 13600 13800 14300 14900 15000 16700  
##      1      3      1      1      1      1      1      1      1
```

Comments : The mean changes significantly more than the median after removing the extreme value (NT\$26,000), whereas the median remains much more stable (robust to outliers). Therefore, the median better describes the typical rent near campus because it is not distorted by extreme values.

Question 3

```
weight <- c(.36,.32,.32)
grade <- c(88, 72, 80)
sum(weight*grade)
```

```
## [1] 80.32
```